

Extreme Programming

Velocity

Velocity in XP is a measure of the amount of work a development team can complete during an iteration (also called a sprint). It is typically measured in **story points**, **ideal days**, or any other consistent unit used to estimate user stories. Velocity helps in **predicting how much work the team can handle in future iterations**, aiding in planning and managing releases effectively.

Steps to Compute Velocity Using the Sustainable Shopping Assistant App Example

1. Break Down Features into User Stories

For the Sustainable Shopping Assistant, we identify key features and break them down into user stories. Here are some examples:

- **User Story 1:** As a user, I want to scan barcodes to get sustainability ratings for products.
 - **User Story 2:** As a user, I want eco-friendly product suggestions based on my shopping habits.
 - **User Story 3:** As a user, I want to see the carbon footprint and ethical sourcing details of products.
 - **User Story 4:** As a user, I want to scan my supermarket bill to analyze my ecological footprint.
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2. Assign Story Points

The development team estimates the **complexity and effort** required for each user story, often using **story points** based on the Fibonacci sequence (1, 2, 3, 5, 8, etc.) or another scale.

- **User Story 1 (Barcode Scanning):** 5 points (moderate complexity—requires camera integration and API calls)
 - **User Story 2 (Eco-friendly Suggestions):** 8 points (complex logic and personalization involved)
 - **User Story 3 (Carbon Footprint Data):** 3 points (simple API integration)
 - **User Story 4 (Bill Scanning & Analysis):** 13 points (OCR processing and data analytics involved)
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3. Complete an Iteration (Sprint)

Assume the first sprint is **2 weeks** long. At the end of the sprint, the team completes:

- **User Story 1 (5 points)**
- **User Story 3 (3 points)**

Total story points completed = 8.

4. Calculate Velocity

Velocity = Total Story Points Completed in the Sprint

- **Sprint 1 Velocity = 8 story points**
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5. Use Velocity for Future Planning

In the next sprint, based on the team's performance, you can plan around completing **8 story points** again. However, velocity might fluctuate initially as the team adjusts, but it stabilizes over time.

Velocity Across Multiple Sprints

Sprint	Stories Completed	Story Points Velocity	
Sprint 1	Barcode Scanning (5), Carbon Footprint (3)	8	8
Sprint 2	Eco-friendly Suggestions (8)	8	8
Sprint 3	Bill Scanning (13)	13	13

Key Points in XP Velocity

- **Sustainable Pace:** Velocity helps maintain a sustainable development pace.
- **Adjustments:** External factors (team changes, complexity underestimation) can impact velocity.
- **Improvement Tool:** It's not just for planning but also for identifying productivity trends.